FREEDOM INTERNATIONAL SCHOOL

WORKSHEET- MCQ (SOLUTIONS)

PHYSICS

THERMODYNAMICS AND KINETIC THEORY

CLASS XI

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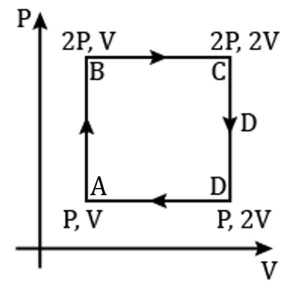
1. During adiabatic compression of a gas, its temperature
   (a) falls (b) remains constant
   (c) rises (d) becomes zero
   Ans: (c) rises
2. An ideal monatomic gas is taken around the cycle ABCDA as shown in the PV diagram. The work done during the cycle is given by
   (a) ½ PV
   (b) PV
   (c) 2 PV
   (d) 4 PV



   Ans: (b) PV
   Work done= Area of the loop ABCDA
   = (2P-P)(2V-V)= PV
3. Air is expanded from 50 litres to 150 litres at 2 atmospheric pressure. The external work done is (1 atmosphere= 1×10^5 N/m^2, 1L= 1×10^{-3} m^3)
   (a) 2×10^{-8} J (b) 2×10^4 J (c) 200 J (d) 2000 J

   Ans: (b) 2×10^4 J

$$W = P \times \Delta V = 2 \times 10^5 \times (150 - 50) \times 10^{-3} = 2 \times 10^4\, J$$

4. $\Delta U + \Delta W = 0$ is valid for
   (a) adiabatic process (b) isothermal process
   (c) isobaric process (d) isochoric process

   Ans: (a) adiabatic process

5. In a cyclic process, work done by the system is
   (a) zero (b) more than the heat given to a system
   (c) equal to heat given to system (d) independent of heat given to system

   Ans: (c) equal to heat given to system

6. The r.m.s speed of a group of 7 gas molecules having speed (6, 4, 2, 0, -2, -4, -6) m/s is
   (a) 1.5 m/s (b) 3.4 m/s (c) 9 m/s (d) 4 m/s

Ans: (d) 4 m/s

$$\text{Mean of squares} = \frac{\text{Sum of squares}}{\text{Number of molecules}} = \frac{112}{7} = 16$$

$$v_{\text{rms}} = \sqrt{\text{Mean of squares}} = \sqrt{16} = 4\,\text{m/s}$$

7. The mean kinetic energy of one mole of a gas per degree of freedom (on the basis of kinetic theory of gases) is

(a) ½ $k_B T$ (b) $\frac{3}{2} k_B T$ (c) $\frac{3}{2}$ RT (d) ½ RT

Ans: (d) ½ RT

8. At which of the following temperatures would the molecules of a gas have twice the average kinetic energy they have at 20 ^{0}C?

(a) 40 ^{0}C (b) 80 ^{0}C (c) 586 ^{0}C (d) 313 ^{0}C

Ans: (d) 313 ^{0}C

$$E \propto T \therefore \frac{E_2}{E_1} = \frac{T_2}{T_1} \Rightarrow \frac{2E_1}{E_1} = \frac{T_2}{(20+273)} \Rightarrow T_2 = 293 \times 2 = 586K = 313^\circ C$$

9. Relation between pressure P and average kinetic energy E per unit volume of a gas is

(a) P= 2E/3 (b) P= E/3 (c) P= 3E/2 (d) P= 3 E

Ans:(a) P= 2E/3

$$KE = \frac{3}{2}PV \Rightarrow P = \frac{2}{3}\frac{KE}{V} = \frac{2}{3}E$$

10. The root mean square velocity of a gas molecule of mass m at a given temperature is proportional to

(a) m^o (b) m (c) $\sqrt{m}$ (d) $m^{-½}$

Ans: (d) $m^{-½}$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** Work and heat are two equivalent forms of energy.
Reason: Work is the transfer of mechanical energy irrespective of temperature difference, whereas heat is the transfer of thermal energy because of temperature difference only.
Ans: A

12. **Assertion:** Air quickly leaking out of a balloon becomes cooler.
Reason: The leaking air undergoes adiabatic expansion.

Ans: A

13. **Assertion:** The specific heat of a gas in an adiabatic process is zero and in an isothermal process is infinite.
Reason: Specific heat of a gas is directly proportional to change of heat in system and inversely proportional to change in temperature.
Ans: A

14. **Assertion:** The total translational kinetic energy of all the molecules of a given mass of an ideal gas is 1.5 times the product of its pressure and its volume.
Reason: The molecules of a gas collide with each other and the velocities of the molecules change due to collision.
Ans: B

15. **Assertion:** The molecules of a monatomic gas have three degrees of freedom.
Reason: The molecules of a diatomic gas have five degrees of freedom.
Ans: B